CAMILA VERA VILLA

GeoAI Researcher — satellite imagery, machine learning & disaster risk

Santiago, Chile · cam.vera@uc.cl · cfvv.github.io · Google Scholar · GitHub: CFVV · LinkedIn

Fulbright Grantee (2024) — PhD applicant, Fall 2027

EDUCATION

M.Sc. in Engineering, Computer Science — Pontificia Universidad Católica de Chile (PUC) 2021 – 2024

Focus: Urban Informatics & Artificial Intelligence · GPA 6.5/7.0 · ANID National Master's Scholarship

- Thesis: *Vector representation learning for urban task applications* — Maximum Distinction (7.0/7.0).

B.Arch. in Architecture, Territorial Urban Project specialization — Universidad de Concepción 2009 – 2014

Rank 5 of 72

B.A. in Education, Elementary Education — Universidad Andrés Bello2012 – 2016

Rank 1 of cohort, Best Student Award · full merit scholarship · completed in parallel with Architecture

RESEARCH EXPERIENCE

GeoAI Researcher - PUC × CIGIDEN (Research Center for Integrated Disaster Risk Management) 2025 – present

Satellite-based exposure modeling for the SIRVAL multi-hazard risk platform, Greater Valparaíso

- Developed territorial change-detection methods on Sentinel-2, Esri Wayback and high-resolution imagery, integrating spectral and geomorphological features.
- Built supervised ML models (Random Forest, Gradient Boosting, Logistic Regression) predicting the emergence and expansion of informal settlements (2018–2025).
- Engineered reproducible Python pipelines (rasterio, geopandas, scikit-learn) from preprocessing to probability-map generation; results feed a public territorial risk platform used by the regional government.

Researcher - Millennium Institute for Foundational Research on Data (IMFD), Cities Group 2021 – present

Project: Data for the study of multidimensional social problems

- Built an end-to-end computer-vision pipeline (Vertex AI) -dataset curation, bounding-box labeling, training and evaluation- for detection of urban features correlated with geolocated crime in Santiago street-view imagery.
- Developed graph neural network models (GCNs) integrating geospatial, textual and imagery signals; led multimodal data collection with data-governance, privacy and ethics review.
- Co-authored peer-reviewed publications; coordinated a cross-functional team (milestones, bi-weekly meetings).

Research Intern - National Institute of Informatics (NII), Kitamoto Lab Apr – Sep 2023

Tokyo, Japan · competitive NII International Internship Program scholarship

- Applied representation learning to urban and geospatial data: siting analysis for solar photovoltaic plants and disaster-risk areas in Japan.
- Presented monthly to Google Japan researchers and, at project close, to renewable-energy and biodiversity experts at Osaka University.

Research Student - National Center for Artificial Intelligence (CENIA), Deep Learning for Vision & Language 2021 – 2024

- Presented work on urban perception and segregation with GNNs at interdisciplinary research workshops; taught Python/AI workshops for high-school teachers and students.

PUBLICATIONS & SOFTWARE

Vera, C., Lucchini, F., Bro, N., Mendoza, M., Löbel, H., Gutiérrez, F., Dimter, J., Cuchacovic, G., Reyes, A., Valdivieso, H., Alvarado, N., & Toro, S. (2022). Learning to cluster urban areas: two competitive approaches and an empirical validation. *EPJ Data Science*, 11, 62. doi:10.1140/epjds/s13688-022-00374-2

Reyes, A., Mendoza, M., **Vera, C.**, Lucchini, F., Dimter, J., Gutiérrez, F., Bro, N., Löbel, H., & Reyes, A. (2024). SpatialCluster: A Python library for urban clustering. *SoftwareX*, 26, 101739. doi:10.1016/j.softx.2024.101739

Software: SpatialCluster — open-source Python library for urban spatial clustering (five methods, evaluation metrics, map visualization). PyPI · contributor (testing, documentation, paper).

CONFERENCE PRESENTATIONS

- KHIPU: Latin American Meeting in Artificial Intelligence, Uruguay, 2023 - poster (travel grant awardee).
- ChileWIC (Chilean Women in Computing), 2022 - finalist & speaker: “Detecting neighborhoods with AI: evaluation of urban clustering methods, Santiago de Chile”.
- Complex Networks Conference, Spain (online), 2021 - poster & talk: “Parametric sensitivity in Graph Neural Networks for urban cluster detection”.
- Conference on Complex Systems, France (online), 2021 - poster; ChileWIC 2021, finalist & poster.

INDUSTRY EXPERIENCE

Data Scientist Engineer - Mercado Libre 2022 – 2025

Shipping Probabilistic Forecasting team (concurrent with research roles; 6-month leave for NII internship)

- Built and productionized probabilistic forecasting models for shipping compliance across 5 Latin American countries (Vertex AI + internal ML platform), with CI/CD, monitoring, and rollback-ready versioning on Kubernetes.
- Led migration of ETL/feature pipelines (Dataflow, BigQuery, GCS) to a data-mesh platform, reducing execution times; peak-season model iterations were deployed to production and replicated by other teams.

Data Scientist - Falabella Retail · Data Analyst - QCart 2020 – 2022

- Multi-country sales analytics, GCP data platform migration (Dataflow/BigQuery), self-serve dashboards; user-behavior analytics for web/mobile products.

Architect & BIM Specialist - luis vidal + architects (Santiago/Dallas) · Carvajal Casariego + Riesco Rivera 2018 – 2020

- BIM modeling of large-scale healthcare and airport infrastructure (Hospital Sótero del Río, 217,630 m²; Dallas–Fort Worth Airport expansion); led public-participation processes. First self-taught Python automations for BIM workflows.

TEACHING EXPERIENCE

Adjunct Instructor - Institute for Mathematical & Computational Engineering and School of Engineering, PUC 2026 – present

- Instructor of record: “Data Science” and “Programming as a Tool for Engineering” (undergraduate).

Teaching Assistant - PUC (graduate programs)2022 – 2026

- Geospatial Data Science (M.Sc. Data Science, 2025-2, 2026-1); Deep Learning (M.Sc. AI, 2022-2); Big Data (MITM, 2022-1); Introduction to Data Science (M.Sc. AI); Pattern Recognition; Introduction to Programming.

Assistant Professor in Data Science - Desafío Latam Academy & Mujeres on IT 2020 – 2022

- Python, machine learning and big data courses; Python-for-data-analysis workshops for women in Chile and Peru.

AWARDS & SCHOLARSHIPS

- Fulbright–ANID Scholarship for PhD studies in the United States — 2024.
- NII International Internship Program scholarship, Japan — 2023 · KHIPU travel grant, Uruguay — 2023.
- ANID National Master's Scholarship — 2021 · PUC Excellence Scholarship for M.Sc. studies (first recipient) — 2021.
- 1st place, NASA Space Apps Challenge Chile (“Planets Near and Far”) — 2019 · 2nd place, Hacking for Humanity (Girls in Tech / AWS) — 2019.
- 1st place, national architecture competition: Memorial to the Victims of Political Repression (built) — 2013 · Full merit scholarship for undergraduate studies in Education (UNAB) — 2012.

OUTREACH & SERVICE

- Host, *Ellas Programan* podcast (IMFD) — two seasons of interviews with women in computing, on Spotify (2022–2024).
- IMFD Gender Commission — co-author of the Institute's gender equity guidelines (2022–2024).
- Volunteer instructor, Technovation Girls Chile & Pre-Ingeniería UC — AI and app-development workshops for school students (2021–2024).
- Post-earthquake building damage surveys, Biobío Region — volunteer surveyor after the 2010 Mw 8.8 Maule earthquake.

TECHNICAL SKILLS

Programming: Python, R, SQL · **ML/DL:** PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face · **Geospatial:** Google Earth Engine, QGIS, GDAL, rasterio, geopandas, OSM

Cloud & MLOps: GCP (Vertex AI, BigQuery, Dataflow), Kubernetes, CI/CD · **Design:** Revit, AutoCAD, Photoshop, Lightroom · **Languages:** Spanish (native), English (professional working proficiency)